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Annealing effect on the structural and optoelectronic properties of Cu-Cr-O thin films deposited by reactive magnetron sputtering using a single CuCr target

Du-Cheng Tsai[1], Erh-Chiang Chen[1], Yen-Lin Huang[2], Fuh-Sheng Shieu[1], Zue-Chin Chang[3,*]

[1]Department of Materials Science and Engineering, National Chung Hsing University, Taichung 402202, Taiwan, Republic of China

[2]Metal Industries Research and Development Centre, Kaohsiung 811225, Taiwan, Republic of China

[3]Department of Mechanical Engineering, National Chin-Yi University of Technology, Taichung 411030, Taiwan, Republic of China

The aim of this study is to explore the structural and optoelectronic properties of Cu-Cr-O thin films when processed by the magnetron sputtering method using a single equimolar CuCr alloy target. These films were then post-annealed in a controlled Ar atmosphere at 500°C to 800°C for 2 h. The as-deposited Cu-Cr-O thin film consisted of an amorphous phase and exhibited extremely poor optoelectronic properties. After annealing was conducted at 500°C, monoclinic CuO and spinel $CuCr_2O_4$ phases were simultaneously formed in the film. Upon increasing the annealing temperature to 600°C, the $CuCr_2O_4$ phase reacted completely with the CuO phase and transformed into the delafossite $CuCrO_2$ phase, possessing optimal optoelectronic performance. It has an electrical resistivity of 41 Ω-cm and a light transmittance of 49.5%, making it suitable for p-type transparent conducting electrodes. A further increase in annealing temperature resulted in larger grains and greater surface roughness and void density, which, in turn, degraded the optoelectronic performance.

Keywords: *Transparent conducting oxide, copper chromium oxide, annealing temperature, optoelectronic properties*

1. Introduction

Thin films with visible-light transmittance above 80% and electrical resistivity below 1×10^{-3} Ω-cm are called transparent conducting oxides (TCOs). The representative TCO thin-film materials include SnO_2, In_2O_3, and ZnO [1]. However, the optoelectronic properties of these TCO thin films are not stable, especially at high ambient temperature. Impurity-doped TCOs, such as Sb_2O_3-doped SnO_2, Al_2O_3-doped ZnO, and SnO_2-doped In_2O_3, are usually preferred to solve this problem [2–4]. Proper dopant content could greatly increase free-electron concentration without affecting visible-light transmittance, resulting in extensive applications in optoelectronic devices. However, these aforementioned TCOs with excellent optoelectronic properties are n-type; p-type TCOs with high visible-light transmittance and low electrical resistivity are also required to develop transparent p–n diode elements in the optoelectronic industry. As a p-type TCO material, $CuCrO_2$ has the advantages of abundant mineral content, low cost, and nontoxic constituent elements, thus showing the greatest potential in optoelectronic applications [5].

Many methods could be used to prepare Cu-Cr-O thin films, including pulsed laser deposition [6], chemical solution deposition [7], solid-state reaction method [8], hydrothermal method [9], chemical vapor deposition [10], and reactive sputter deposition [11–14]. Among them, reactive sputter deposition is one of the most frequently used methods for growing thin films because it can provide highly uniform thickness in large-area depositions. The sputtering deposition techniques for Cu-Cr-O thin films often utilize a sintered ceramic target as the source [11]. Ceramic targets, however, require a more complicated manufacturing pro-

* E-mail: chang@ncut.edu.tw

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cess than metal targets, and thus, their cost is extremely high overall. In addition, they are prone to cracking due to the high temperature generated by the plasma during the sputtering process. By using a dual-target sputtering system in an oxygen-incorporation atmosphere, Cu and Cr metal targets could be used to directly deposit Cu-Cr-O thin film [14]. This method eliminates the need to prepare Cu-Cr-O powder, which is time and resource intensive. In addition, the conductivity of the metal target is higher than that of the Cu-Cr-O target; this facilitates the use of DC sputtering techniques. Nevertheless, the compositional uniformity of dual-target sputtering is significantly poor under large-area deposition due to the difference in the positions of both targets. A single CuCr alloy target could be used for sputtering in an oxygen atmosphere to resolve the problems mentioned above. Thus, this method was adopted in the present work to synthesize Cu-Cr-O thin film. The actual optoelectronic performance of this method, however, remains to be verified in this study.

On the other hand, the fabrication of as-deposited Cu-Cr-O thin films is substantially dependent on good control of the post-annealing temperature as the as-sputtered thin films generally reveal poor crystallinity. Hence, this study focused on optimizing the quality of thin films by altering the post-annealing temperature. The influence of phase composition and microstructural variation on the optoelectronic characterization of the Cu-Cr-O thin film was also explored. Finally, the feasibility of the method was evaluated.

2. Materials and experimental methods

High-vacuum DC magnetron sputter deposition equipment was employed to grow Cu-Cr-O thin films on fused silica substrates. The target used was a CuCr alloy target with diameters of 4 in, and its atomic ratio was 1:1. By using a turbo molecular pump with a rotary pump, the system achieved an ultimate vacuum of 3.0×10^{-6} Torr. The target and the substrate had a working distance of about 90 mm. Before deposition, the substrates were sequentially cleaned and rinsed with ethanol and distilled water in an ultrasonic bath. Film deposition was carried out at a CuCr-target power of 200 W, a substrate temperature of room temperature, and an $O_2/(O_2 + Ar)$ flow rate ratio of 10%. The working pressure was 5.0×10^{-3} Torr at this moment. The sputtering time was set to 15 min to obtain a film thickness of 330 nm. Following deposition, the specimens were post-annealed in a tube furnace under an Ar flow rate of 400 sccm (standard cubic centimeter per minute). The oxygen partial pressure in the tube furnace was kept below 7.0×10^{-5} atm. The annealing temperature was ramped from room temperature to a temperature range of 500°C–800°C (equivalent to 773 K—1073 K) at a rate of 5°C/min for 2 hours, and then cooled down to room temperature at the same rate.

Electron spectroscopy for chemical analysis (ESCA, PHI 5000 VersaProbe) was used to determine the chemical composition of Cu-Cr-O films. X-ray diffraction (XRD, Mac-Science MXP3) equipped with Cu Kα radiation was employed to examine the phase structure of these films, with a scanning speed of 1°/min. Field-emission scanning electron microscopy (FESEM, JEOL JSM-6700F) and field-emission transmission electron microscopy (FETEM, JEOL JEM-1200EXII) were utilized to observe the microstructure of Cu-Cr-O films. The atomic force microscope (AFM, Seiko SPA400) was utilized to examine the surface topography of Cu-Cr-O films. Ultraviolet/visible spectrophotometry (Hitachi U3010) was applied to gauge the transmittance of Cu-Cr-O films. The Hall effect measurement system (Accent HL5500PC) was operated to measure the electrical property of Cu-Cr-O films.

3. Results and discussion

3.1. Chemical composition

The variation in the chemical composition of Cu-Cr-O films with annealing temperature is revealed in Table 1, which shows two significant findings. First, the ratio of Cu to Cr in these films was measured to be about 2, although an equimolar CuCr target was used in this study. The higher

Table 1. The elemental composition of Cu-Cr-O films as a function of annealing temperature

Annealing temperature (°C)	Cu (at.%)	Cr (at.%)	O (at.%)	Cu/Cr ratio
As-deposited	28.41	14.05	57.54	2.02
500	31.69	15.95	52.36	1.99
600	31.20	16.24	52.55	1.92
700	31.23	15.68	53.09	1.99
800	31.81	15.68	52.51	2.03

Cu content in the films may be due to the difference in sputtering yields between Cu and Cr [15]. Second, only marginal variation was observed in the chemical composition with an increment in annealing temperature. The evaporation of these constituent elements did not occur in the films even when annealed at temperatures up to 800°C.

3.2. Phase transformation

Figure 1 shows the XRD patterns of the Cu-Cr-O films with different post-annealing temperatures. No visible diffraction peaks appeared in the as-deposited film, indicating that it was amorphous. As the annealing temperature was heightened to 500°C, the film began to crystallize, mainly forming spinel $CuCr_2O_4$ and monoclinic CuO phases. From the perspective of elemental composition, the higher the copper content, the more preferentially the copper-rich phase is formed. However, in terms of thermodynamics, the Gibbs free energy of the $CuCr_2O_4$ phase is relatively negative

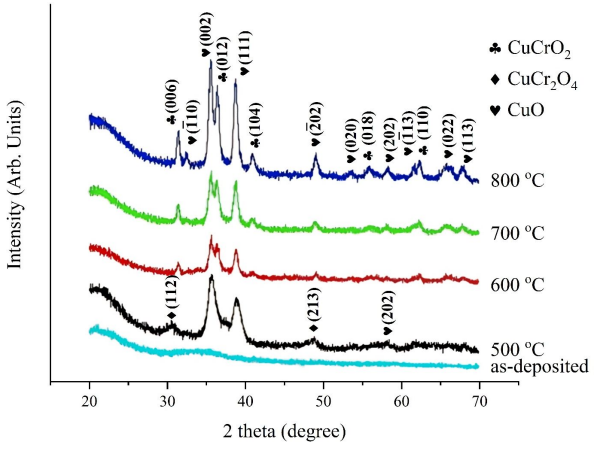


Fig. 1. XRD patterns of Cu-Cr-O films with different post-annealing temperatures

at lower temperatures, indicating that the generation of $CuCr_2O_4$ phase is thermodynamically favorable [16]. When the annealing temperature attained 600°C, the spinel $CuCr_2O_4$ phase disappeared, and it was replaced by the delafossite $CuCrO_2$ phase, because increasing the annealing temperature rapidly reduces the Gibbs free energy of the $CuCrO_2$ phase even lower than that of the $CuCr_2O_4$ phase, thereby promoting the reaction of $CuCr_2O_4$ with CuO to form $CuCrO_2$. These relevant reaction equations and thermodynamic data can be expressed as follows:

$$1/2Cu_2O_{(s)} + 1/4O_{2(g)} \rightarrow CuO_{(s)} \quad (1)$$

$$\Delta G_1 = \Delta G_1^\circ + RTln(pO_2) = -65222.25 + 46.955T - RTln(pO_2) \quad (2)$$

$$CuO_{(s)} + Cr_2O_{3(s)} \rightarrow CuCr_2O_{4(s)} \quad (3)$$

$$\Delta G_2 = \Delta G_2^\circ + RTln(pO_2) = 3667 - 15.85T \quad (4)$$

$$CuCr_2O_{4(s)} + CuO_{(s)} \rightarrow 2CuCrO_{2(s)} + 1/2O_{2(g)} \quad (5)$$

$$\Delta G_3 = \Delta G_3^\circ + RTln(pO_2) = 79512.5 - 68.15T + RTln(pO_2) \quad (6)$$

where ΔG and ΔG° are the variations of the Gibbs free energy and the standard Gibbs free energy, respectively. R is the gas constant, T is the absolute temperature, and pO_2 is the oxygen partial pressure. In these equations, the subscript "(s)" is utilized to indicate substances in the solid state, while the subscript "(g)" represents substances in

the gaseous state. From the above equations, when the pO_2 is 7.0×10^{-5} atm, the ΔG can be calculated, and the variation of ΔG with temperature can be drawn, as shown in Figure 2. When the annealing temperature exceeds 600°C, it is found that $\Delta G_3 < \Delta G_2 < \Delta G_1$, which means that the $CuCrO_2$ phase becomes a thermodynamically favorable phase. As the annealing temperature was heightened to 800°C, the diffraction peaks became stronger and sharper, implying grain growth in the $CuCrO_2$ and CuO phases.

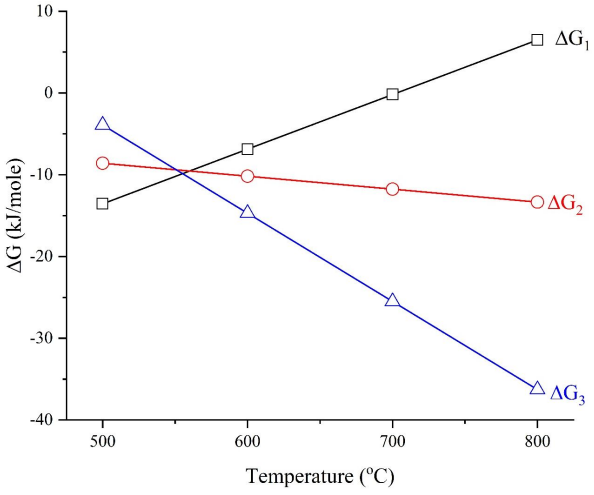


Fig. 2. The ΔG of reactions 2, 4, and 6 with different annealing temperatures

3.3. Microstructure

Figure 3 shows the SEM image of the Cu-Cr-O films with different post-annealing temperatures. The as-deposited film was composed of nanoparticles with size ranging from approximately 4 nm to 10 nm (Figure 3a). An annealing procedure at 500°C generated larger particles, about 30–40 nm in size, with an agglomerate surface characteristic that left voids inside the film (Figure 3b). With the further increase in annealing temperature to 800°C, the film grains grew significantly, and the particles size increased to hundreds of nanometers (Figures 3c–3e). While higher temperature increases particle size by enhancing atomic diffusion, it also enlarges the void by causing atoms to coalesce near the void edges due to capillary forces. Thus, whether the film properties obtained by high-temperature annealing are better is worth exploring. Figure 4 shows the AFM images of the Cu-Cr-O films with different post-annealing temperatures. These results were in agreement with the FESEM observations. The high heat induces coalescence of adjacent grains, resulting in a rapid increase in RMS roughness from 0.75 nm to 20.01 nm with increasing annealing temperature. Figure 5 shows TEM images with the selected area diffraction (SAD) patterns of the Cu-Cr-O films with different post-annealing temperatures. The 500°C-annealed film consisted of equiaxed CuO and $CuCr_2O_4$ grains (Figure 5a). The grains and void size increased along the film growth direction. The thickness-dependent change in grain size is a common phenomenon for annealed growth. When the annealing temperature was increased to 600°C, the film exhibited increased void density and grain growth (Figure 5b). In this situation, the $CuCr_2O_4$ reacted with CuO to form $CuCrO_2$ until $CuCr_2O_4$ was depleted. However, no distinct two-phase structure in this film could be observed, probably due to insufficient atom diffusion at the present annealing temperature. When the annealing temperature was continued to increase to 800°C, these grains and voids also continued to grow (Figure 5c). Visible phase segregation was present in the microstructure, as confirmed by EDS analysis (not shown). The above findings were in good agreement with the XRD analysis.

3.4. Optical measurements

Figure 6a shows the light transmittance of the Cu-Cr-O films with different post-annealing temperatures. The as-deposited film only exhibited a very low transmittance of 7.9% in a wavelength region of 500–900 nm. Increasing the annealing temperature to 500°C resulted in sharper fundamental absorption edges and increased transmittance up to 31.4%, which was caused by improved crystal quality. These CuO and $CuCr_2O_4$ with narrow band gap and high photo response in the film were responsible for this result [17, 18]. At an annealing temperature of 600°C, the film achieved the highest transmittance of 49.5%. The increase

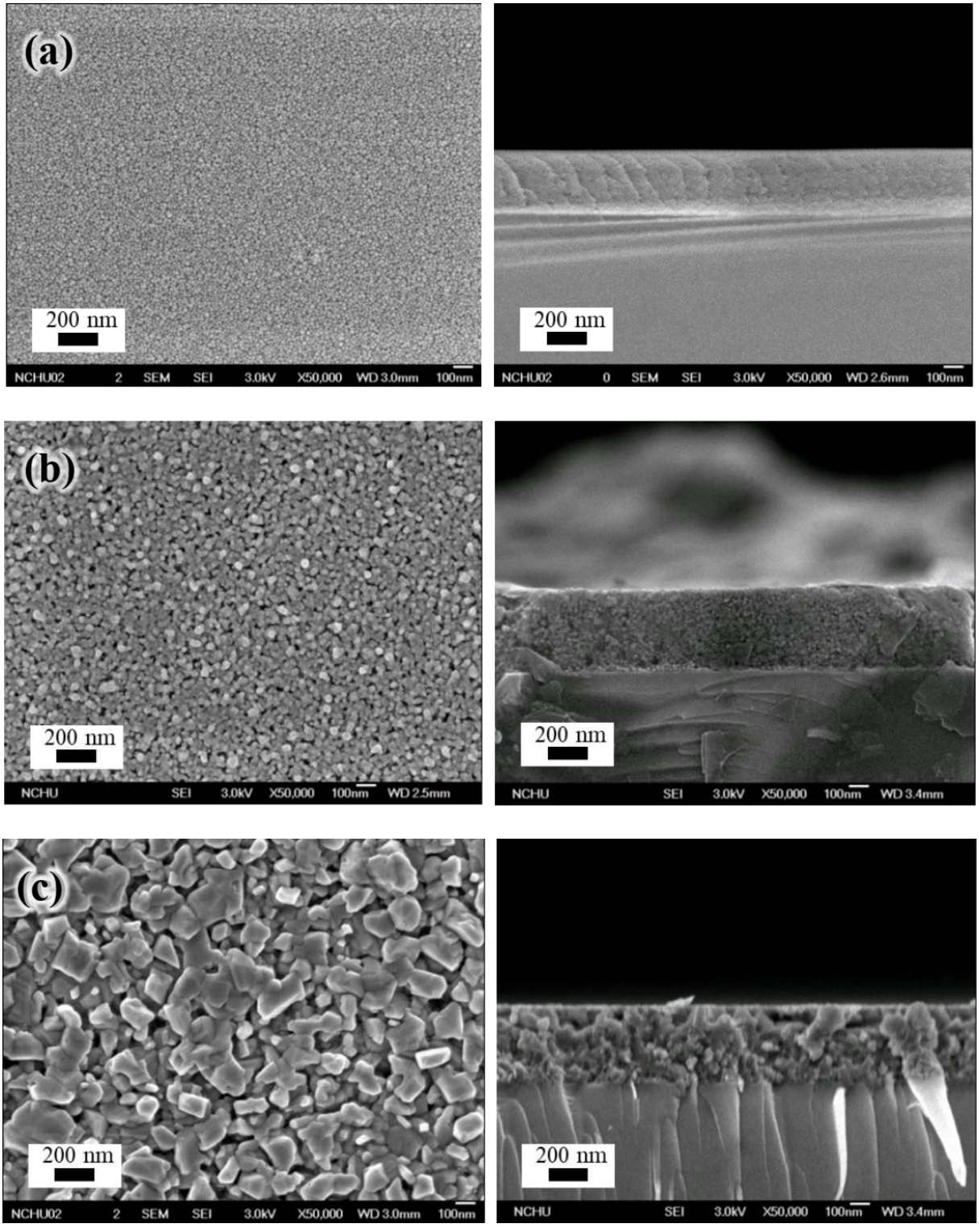


Fig. 3. Cross-section and plane-view SEM images of Cu-Cr-O films with different post-annealing temperatures: (a) as-deposited, (b) 500°C, (c) 600°C, (d) 700°C, and (e) 800°C

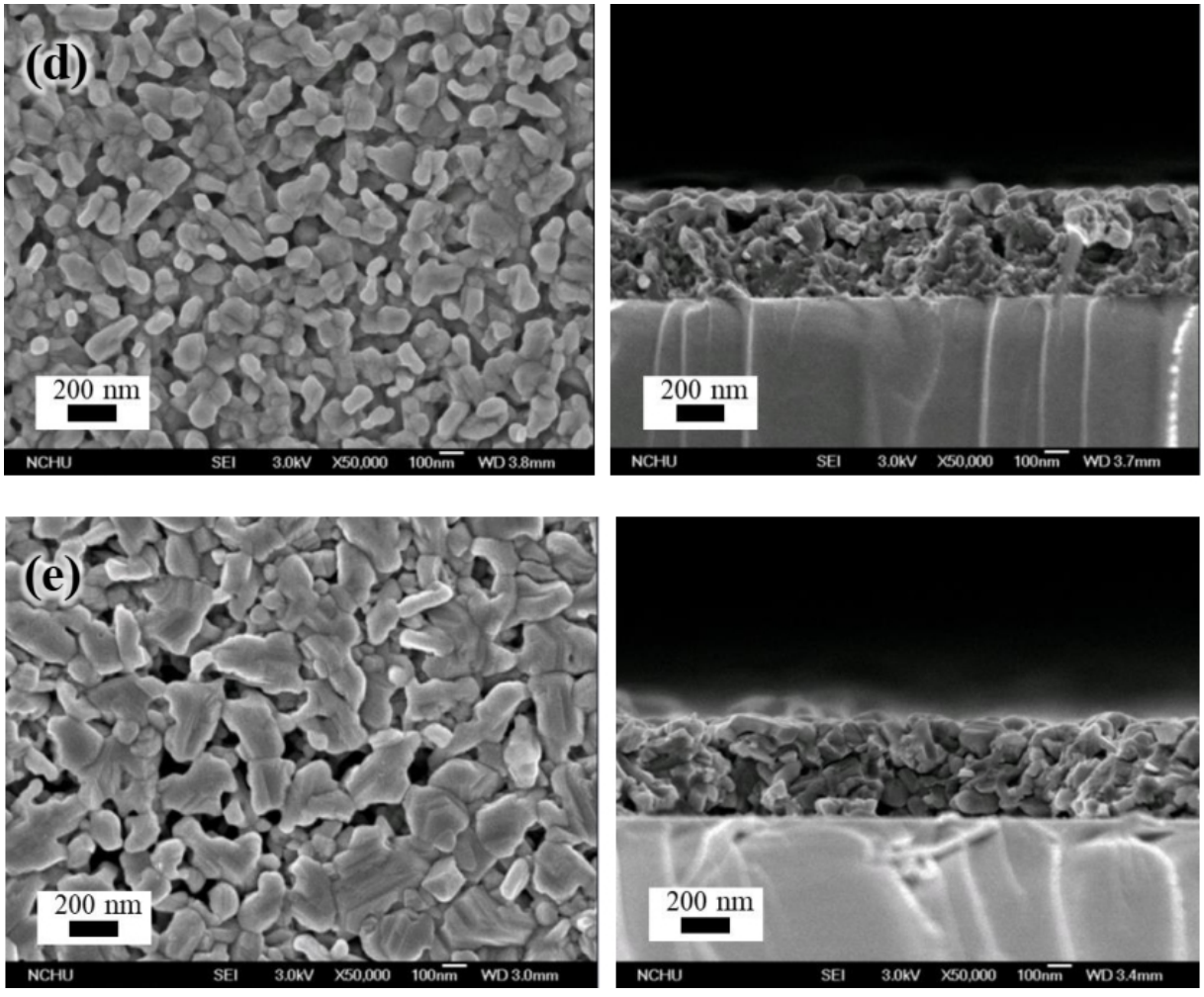


Fig. 3. ...continued

in grain size and the replacement of the $CuCr_2O_4$ phase by the $CuCrO_2$ phase with a wide band gap were the major factors. When the annealing temperature was 800 °C, however, the transmission of the film decreased to 36.7%, which was associated with higher roughness, increased void defects, and larger CuO grains. The part of incident light was redirected to diffuse scattering instead of specular scattering because of the increment in surface roughness and void defects [19]. The as-deposited film was characterized by a gentle absorption edge, implying its high defect density in film. As annealing temperature increased, the absorption edge became sharper, which was attributed to crystallization and grain growth. The band gap of the Cu-Cr-O films was measured by a linear extrapolation on the x-axis of plot of the $(\alpha h\nu)^2$ vs. photon energy ($h\nu$), as shown in Figure 6b. The band gap energy values of the as-deposited, 500°C-annealed, 600°C-annealed, 700°C-annealed, and 800°C-annealed films were found to be 1.25, 1.78, 1.98, 1.96, and 1.84 eV, respectively. The low band gap of the deposited films was also associated with a high defect density. The initial increase of band gap with was dependent on the formation and growth of these crystal phases. This subsequent decrease of band gap may be related to a reduction in carrier concentration, known as the Burstein–Moss effect [20, 21].

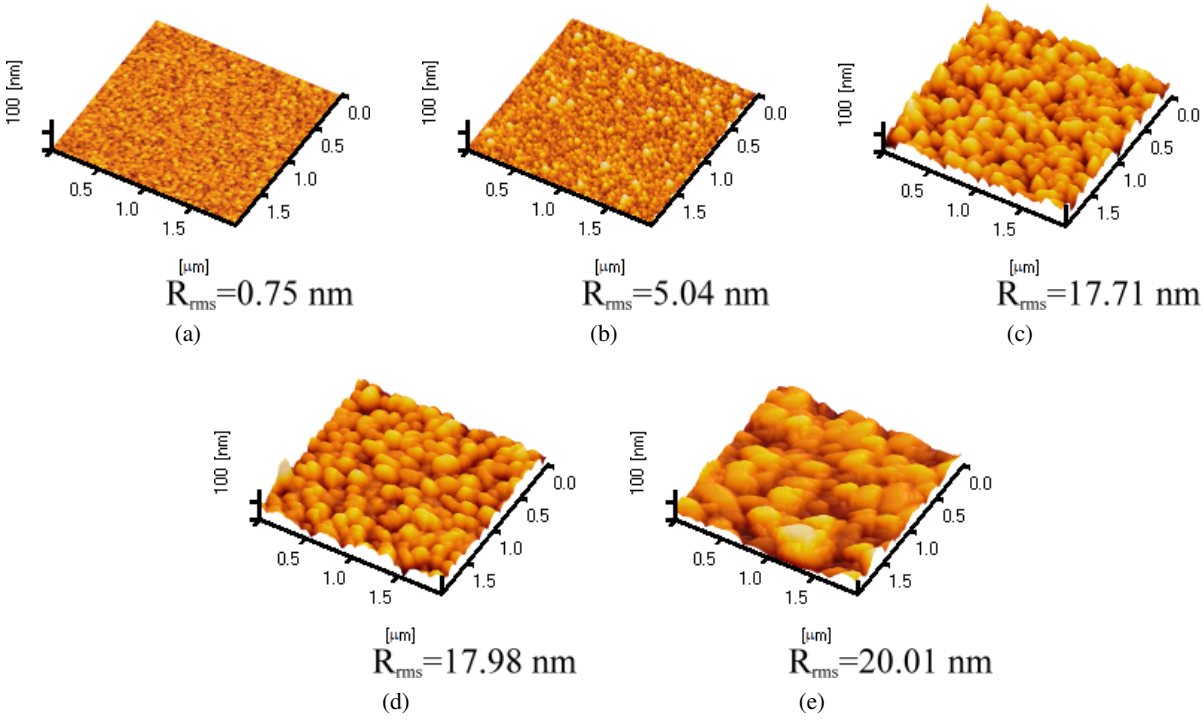


Fig. 4. AFM images of Cu-Cr-O films with different post-annealing temperatures: (a) as-deposited, (b) 500°C, (c) 600°C, (d) 700°C, and (e) 800°C

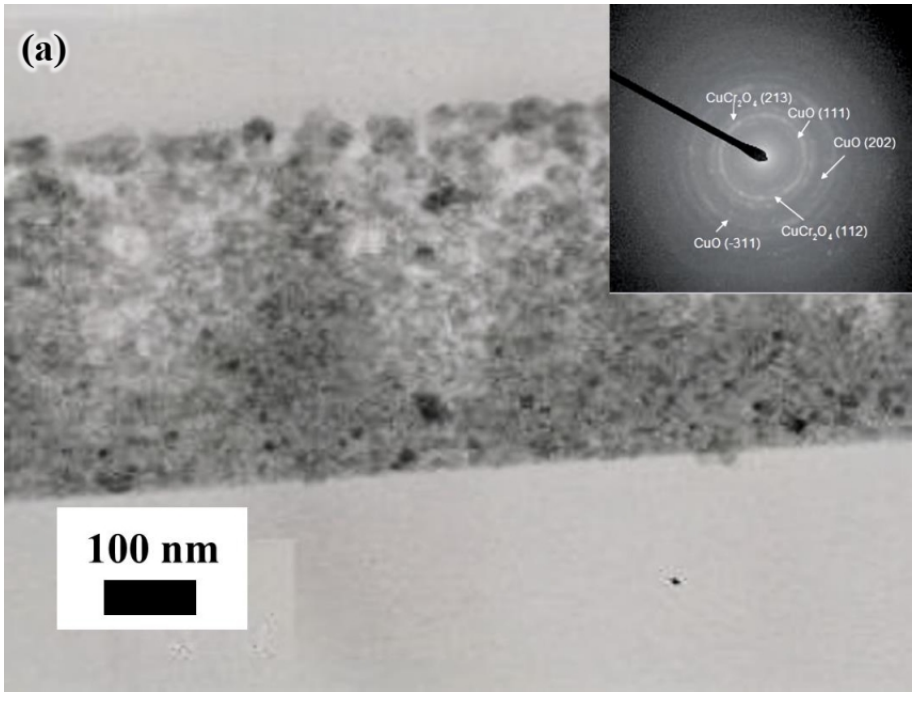


Fig. 5. TEM images with corresponding SAD patterns (insets) of Cu-Cr-O films with different post-annealing temperatures: (a) 500°C, (b) 600°C, and (c) 800°C

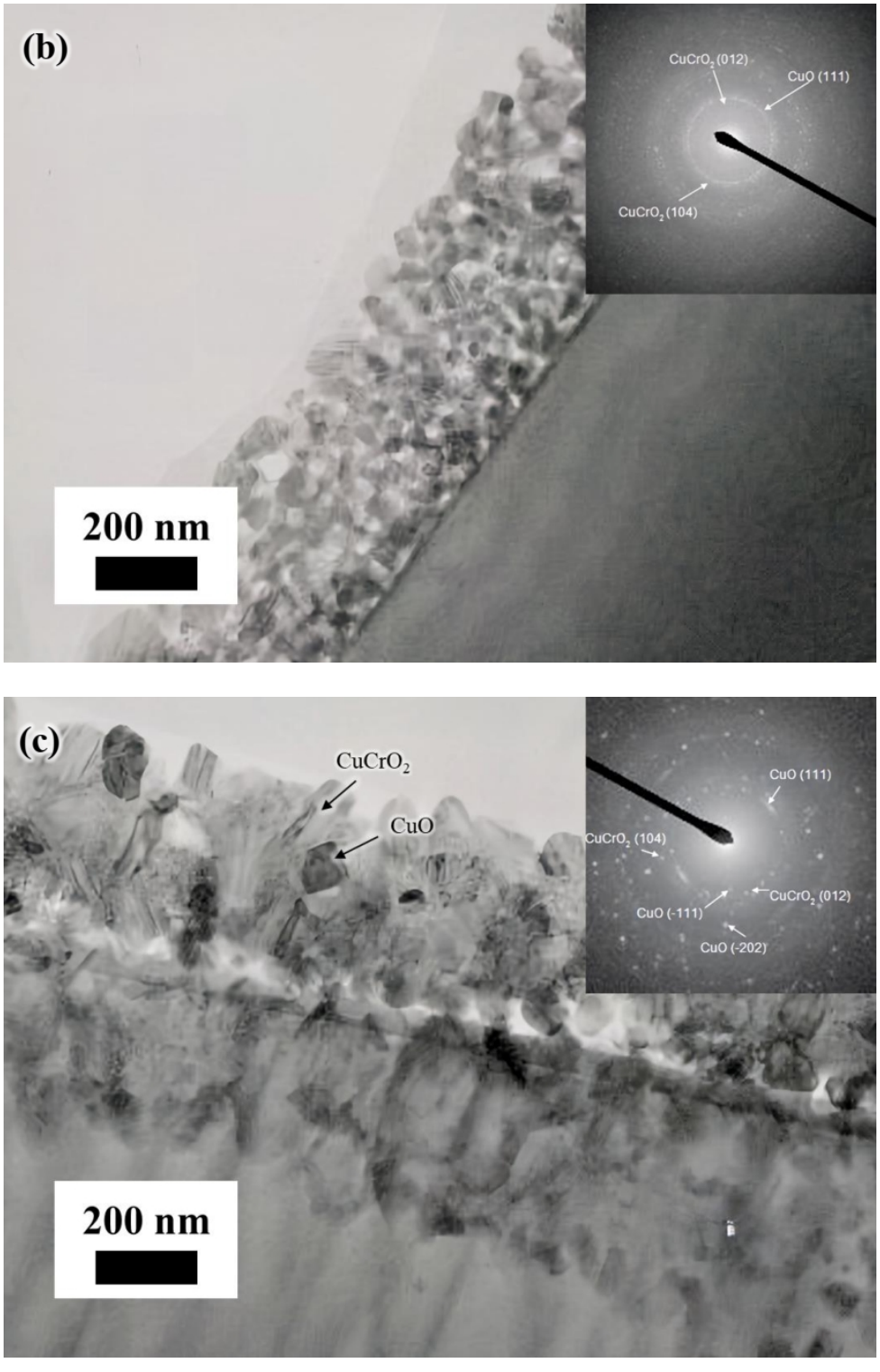


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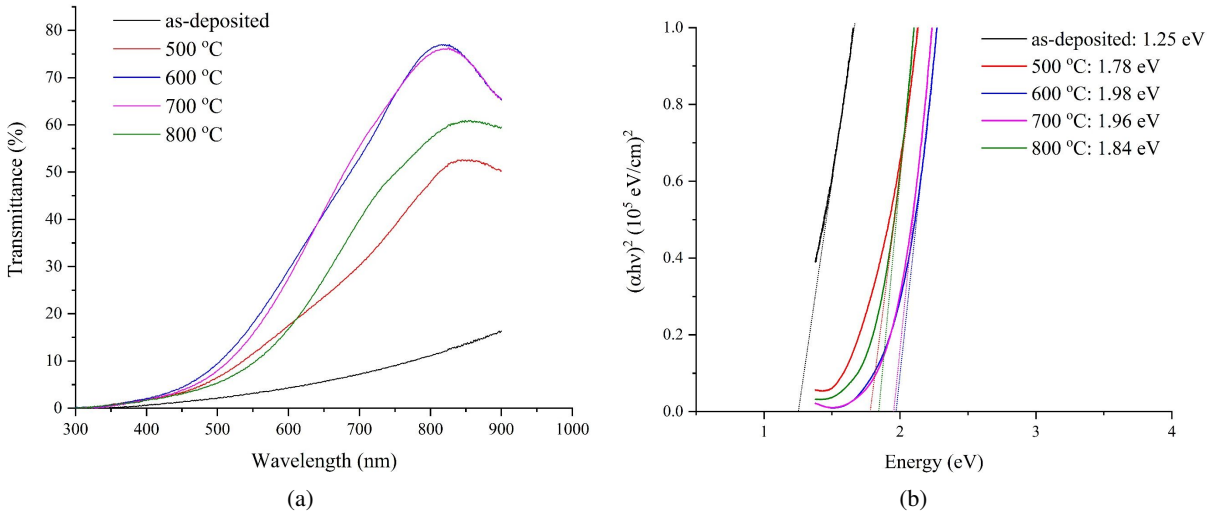


Fig. 6. (a) Light transmittance of Cu-Cr-O films with different post-annealing temperatures. (b) variation of $(\alpha h\nu)^2$ versus $h\nu$

3.5. Electrical measurements

Table 2 shows the Hall effect measurement results for the Cu-Cr-O films with different post-annealing temperatures. Due to the positive Hall coefficients measured in the Cu-Cr-O films, the holes were confirmed to be the major carriers of these films, indicating a p-type semiconductor behavior. A considerable variation was found in the electrical property of CuO, $CuCr_2O_4$, and $CuCrO_2$ films studied by different authors, likely due to differences in the films' composition and structure that contributed to this inability to fully explain electrical behavior. The as-deposited film had a very high electrical resistivity that could not be measured by a Hall system, hence it was omitted from Table 2. After annealing was conducted at 500°C, the films exhibited measurable electrical resistivity. When the annealing temperature attained 600°C, the carrier concentration and Hall mobility of the film increased. $CuCrO_2$ is generally known to have lower electrical resistivity than $CuCr_2O_4$ [22–25]. Coupled with the larger grain size, the substitution of $CuCr_2O_4$phase with the $CuCrO_2$ phase was thought to be responsible for the superior electrical performance. As the annealing temperature was further increased, however, the electrical property of the film deteriorated, mainly due to the decrease in carrier concentration. Yoshida et al. have confirmed that the major source of conductive carriers in Cu-Cr-O p-type semiconductors was interstitial O and Cu vacancies [26, 27]. In addition, Banerjee et al. found that electrical resistivity was reduced by extra O atoms present at interstitial sites and grain boundaries [28]. In the present study, while annealing could increase the grain size, it also reduced some point defects that contributed to hole concentration, such as such as interstitial O and Cu vacancies, leading to increased electrical resistivity. The values of Hall mobility showed scattering variations with increasing annealing temperature from 600°C to 800°C. The increase in Hall mobility originated from the decrease in grain boundary scattering and point defect scattering due to the increase in grain size [29]. However, the larger surface roughness and void density at higher annealing temperature facilitated the chemisorption of nitrogen, oxygen, and water vapor into the film, thus increasing barrier height and degrading Hall mobility [30]. Accordingly, the film annealed at 600°C had the lowest electrical resistivity of 41 Ω-cm.

Based on the above results, although 600°C-annealed film has the optimal optoelectronic performance, its phase composition contains a large amount of CuO phase with narrow energy gap, resulting in poor light transmittance compared with

Table 2. The carrier concentration, Hall mobility, and electrical resistivity of Cu-Cr-O films as a function of annealing temperature

Annealing temperature (°C)	Carrier concentration ($\times 10^{17}$ cm^{-3})	Hall Mobility (cm^2/V-s)	Electrical resistivity (Ω-cm)
500	2.81	0.121	183
600	4.05	0.373	41
700	3.25	0.223	86
800	0.43	0.268	540

other published papers. The ESCA result shows that the Cu content in film is about twice the Cr content. The composition difference between the film and the target caused by the magnetron sputtering method is too large to be ignored, which will be an unfavorable factor that must be considered and eliminated by adjusting the target composition in the further study of single-target magnetron sputtering in the future.

4. Conclusion

This study investigated the fabrication of transparent conductive Cu-Cr-O films on fused silica substrates through DC reactive magnetron sputtering using a single equimolar CuCr alloy target in conjunction with a post-annealing process. Annealing at 500°C transformed the phase structure of the film from amorphous to monoclinic CuO and spinel $CuCr_2O_4$. Further annealing at 600°C depleted the $CuCr_2O_4$ phase to generate the delafossite $CuCrO_2$ phase. As the annealing temperature increased, the microstructure of the films converted from a compact nanograins feature to large agglomerated grains with numerous voids, thereby roughening the surface. Accordingly, grain growth combined with the formation of the $CuCrO_2$ phase led to the optimized optoelectronic performance of the 600°C-annealed films. However, some subsequent effects associated with grain growth adversely affected the optoelectronic properties. As the annealing temperature continued to increase, the grains also grew continuously, along with the surface roughness and void density. The increased grain size resulted in the reduction in the defects contribution to the hole concentration. The great surface roughness and void density provided electrical barriers for hole transport, thereby lowering the Hall mobility. They also redistributed some of the incident light into diffuse scattering, resulting in significant light-scattering losses.

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Conflict of interest

On behalf of all authors, the corresponding author states that there is no conflict of interest.

References

[1] Jaffray W, Saha S, Shalaev VM, Boltasseva A, Ferrera M. Transparent conducting oxides: from all-dielectric plasmonics to a new paradigm in integrated photonics. Adv Opt Photonics. 2022;14: 148–208.

[2] Gong W, Wang G, Gong Y, Zhao L, Mo L, Diao H, et al. Investigation of In_2O_3:SnO_2 films with different doping ratio and application as transparent conducting electrode in silicon heterojunction solar cell. Sol Energy Mater Sol Cells. 2022;234: 111404.

[3] Joo YH, Um DS, Kim CI. Tunable physical properties of Al-doped ZnO thin films by O_2 and Ar plasma treatments. Mater Res Express. 2021;8: 126402.

[4] Jang KS, Lee JW, Kim J, Yoo SI. Electrical and optical properties of Sb-doped SnO_2 thin films fabricated by pulsed laser deposition. J Korean Ceram Soc. [n.d.];51: 43–50.

[5] Marquardt MA, Ashmore NA, Cann DP. Crystal chemistry and electrical properties of the delafossite structure. Thin Solid Films. 2006;496: 146–156.

[6] Li D, Fang X, Zhao A, Deng Z, Dong W, Tao R. Physical properties of $CuCrO_2$ films prepared by pulsed laser deposition. Vacuum. 2010;84: 851–856.

[7] Lin SH, Yeh RH, Chu C, Yu RS. Effects of Mg doping on structural and optoelectronic properties of p-type semiconductor $CuCrO_2$ thin films. Materials Science Semiconductor Processing. 2022;139: 106346.

[8] Li W, Cheng H. Cu–Cr–O nanocomposites: synthesis and characterization as catalysts for solid state propellants. Solid State Sci. 2007;9: 750–755.

[9] Huang RJ, Sakthinathan S, Chiu TW, Dong C. Hydrothermal synthesis of high surface area $CuCrO_2$ for H2 production by methanol steam reforming. RSC Adv. 2021;11: 12607–12613.

[10] Popa PL, Crêpellière J, Leturcq R, Lenoble D. Electrical and optical properties of Cu–Cr–O thin films fabricated by chemical vapour deposition. Thin Solid Films. 2016;612: 194–201.

[11] Barnabé A, Thimont Y, Lalanne M, Presmanes L, Tailhades P. p-Type conducting transparent characteristics of delafossite Mg-doped CuCrO2 thin films prepared by RF-sputtering. J Mater Chem C. 2015;3: 6012–6024.

[12] Sun CH, Tsai DC, Chang ZC, Chen EC, Shieu FS. Effects of annealing time on the structural and optoelectronic properties of p-type conductive transparent Cu–Cr–O films. J Mater Sci Mater Electron. 2016;27: 9740–9747.

[13] Sun CH, Tsai DC, Chang ZC, Chen EC, Shieu FS. Structural, optical, and electrical properties of conducting p-type transparent Cu–Cr–O thin films. Ceram Int. 2016;42: 13697–13703.

[14] Tsai DC, Chang ZC, Kuo BH, Chen CM, Chen EC, Shieu FS. Influence of chemical composition on phase transformation and optoelectronic properties of Cu–Cr–O thin films by reactive magnetron sputtering. J Mater Res Technol. 2019;8: 690–696.

[15] Zheng B, Wu Z, Cui S, Xiao S, Liu L, Lin H, et al. Discharge and deposition characteristics of high-power impulse magnetron sputtering using various target materials. IEEE Trans Plasma Sci. 2019;47:93–198.

[16] Jacob KT, Kale GM, Iyengar GNK. Oxygen potentials, Gibbs' energies and phase relations in the Cu-Cr-O system. J Mater Sci. 1986;21: 2753–2758.

[17] Hu ZQ, Qin Y, Zhou HR, Kang J, Zhai SR, Gao H. Preparation and photoelectric properties of $CuCr_2O_4$ nanopowders. Adv Mater Res. 2011;284–286: 974–979.

[18] An Y, Li Z, Shen J. Enhanced visible light response of CuO nanoparticle-modified sodium titanate nanotube heterojunction thin films. Mater Lett. 2012;74: 65–67.

[19] Coriand L. *Roughness, Wetting, and Optical Properties of Functional Surfaces*. Stuttgart; 2014.

[20] Tsai DC, Chen FK, Chang ZC, Kuo BH, Chen EC, Huang YL, et al. Effect of Al_2O_3-doping on the structure and optoelectronic characteristics of MgZnO thin film prepared by RF magnetron sputtering. Appl Phys A. 2021;127: 571.

[21] Tsai DC, Chen FK, Chang ZC, Kuo BH, Chen EC, Huang YL, et al. Influence of substrate temperature on the structural and optoelectronic properties of ZnMgO:Al coatings deposited by radio frequency magnetron sputtering. J Ceram Soc Jpn. 2021;129: 714–719.

[22] De KS, Ghose J, Murthy KSRC. Electrical properties of the $CuCr_2O_4$ spinel catalyst. J Solid State Chem. 1982;43: 261–266.

[23] Slade PG. *Electrical Contacts: Principles and Applications*. 2nd ed. Boca Raton, FL: CRC Press; 2017.

[24] Ogwu AA, Darma TH, Bouquerel E. Electrical resistivity of copper oxide thin films prepared by reactive magnetron sputtering. J Achiev Mater Manuf Eng. 2007;172–177.

[25] Banerjee AN, Chattopadhyay KK. Recent developments in the emerging field of crystalline p-type transparent conducting oxide thin films. Prog Cryst Growth Charact Mater. 2005;50: 52–105.

[26] Katayama-Yoshida H, Koyanagi T, Funashima H, Harima H, Yanase A. Engineering of nested Fermi surface and transparent conducting p-type delafossite $CuAlO_2$: possible lattice instability or transparent superconductivity? Solid State Commun. 2003;126: 135–139.

[27] Katayama-Yoshida H, Sato K, Kizaki H, Funashima H, Hamada I, Fukushima T, et al. Ab initio materials design for transparent-conducting-oxide-based new-functional materials. Appl Phys A. 2007;89: 19–27.

[28] Banerjee AN, Ghosh CK, Chattopadhyay KK. Effect of excess oxygen on the electrical properties of transparent p-type conducting CuAlO2+x thin films. Sol Energy Mater Sol Cells. 2005;89: 75–83.

[29] Yu RS, Tasi CP. Structure, composition and properties of p-type $CuCrO_2$ thin films. Ceram Int. 2014;40: 8211–8217.

[30] Lu JG, Ye ZZ, Zeng YJ, Zhu LP, Wang L, Yuan J, et al. Structural, optical, and electrical properties of (Zn,Al)O films over a wide range of compositions. J Appl Phys. 2006;100: 073714.

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